

Mathematical Methods — Units 3 & 4

Differentiation · AI-Marked Submission

18 / 20

Overall: 90% — outstanding.

Strong, well-structured methods throughout, Aisha. Two small final-line slips (Q3 constant, Q5 denominator) cost the last marks. Tidy those and you're at full marks. Great consistency on the chain and product rules.

1. Differentiate $y = x^3 - 4x^2 + 7x$. 4 / 4
Your working: $dy/dx = 3x^2 - 8x + 7$ ■
Feedback: Perfect — clean and correct.
2. Find dy/dx for $y = (2x + 1)^5$. 4 / 4
Your working: $10(2x + 1)^4$ ■
Feedback: Chain rule applied correctly.
3. Find the tangent to $y = x^2 - 3x$ where $x = 2$. 3 / 4
Your working: gradient = 1 at (2, -2), so $y = x - 2$ ■
Feedback: Method correct — slip in the constant. Tangent is $y = x - 4$.
4. Differentiate $y = x e^x$ and find the stationary point. 4 / 4
Your working: $dy/dx = e^x(1 + x)$, stationary at $x = -1$ ■
Feedback: Excellent, fully correct.
5. Differentiate $y = \ln(3x^2 + 1)$. 3 / 4
Your working: $dy/dx = 6x / (3x^2)$ ■
Feedback: Don't drop the +1 in the denominator: $6x / (3x^2 + 1)$.